BSgenome

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Dependencies

This document has the following dependencies:

```
library(BSgenome)
library(BSgenome.Scerevisiae.UCSC.sacCer2)
```

Use the following commands to install these packages in R.

```
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("BSgenome", "BSgenome.Scerevisiae.UCSC.sacCer2"))
```

Corrections

Improvements and corrections to this document can be submitted on its GitHub in its repository.

Overview

The *BSgenome* package contains infrastructure for representing genome sequences in Bioconductor.

Other Resources

Genomes

The *BSgenome* package provides support for genomes. In Bioconductor, we have special classes for genomes, because the chromosomes can get really big. For example, the human genome takes up several GB of memory.

The `available.genomes()` function lists which genomes are currently available from from Bioconductor (it is possible to make your own genome package). Note that there are several so-called "masked" genomes, where some parts of the genome are masked. We will avoid this subject for now.

Let us load the latest yeast genome

```
available.genomes()
```

```
##  [1] "BSgenome.Alyrata.JGI.v1"
##  [2] "BSgenome.Amellifera.BeeBase.assembly4"
##  [3] "BSgenome.Amellifera.UCSC.apiMel2"
##  [4] "BSgenome.Amellifera.UCSC.apiMel2.masked"
##  [5] "BSgenome.Athaliana.TAIR.04232008"
##  [6] "BSgenome.Athaliana.TAIR.TAIR9"
##  [7] "BSgenome.Btaurus.UCSC.bosTau3"
##  [8] "BSgenome.Btaurus.UCSC.bosTau3.masked"
##  [9] "BSgenome.Btaurus.UCSC.bosTau4"
## [10] "BSgenome.Btaurus.UCSC.bosTau4.masked"
## [11] "BSgenome.Btaurus.UCSC.bosTau6"
## [12] "BSgenome.Btaurus.UCSC.bosTau6.masked"
## [13] "BSgenome.Btaurus.UCSC.bosTau8"
## [14] "BSgenome.Celegans.UCSC.ce10"
## [15] "BSgenome.Celegans.UCSC.ce2"
## [16] "BSgenome.Celegans.UCSC.ce6"
## [17] "BSgenome.Cfamiliaris.UCSC.canFam2"
## [18] "BSgenome.Cfamiliaris.UCSC.canFam2.masked"
## [19] "BSgenome.Cfamiliaris.UCSC.canFam3"
## [20] "BSgenome.Cfamiliaris.UCSC.canFam3.masked"
## [21] "BSgenome.Dmelanogaster.UCSC.dm2"
## [22] "BSgenome.Dmelanogaster.UCSC.dm2.masked"
## [23] "BSgenome.Dmelanogaster.UCSC.dm3"
## [24] "BSgenome.Dmelanogaster.UCSC.dm3.masked"
## [25] "BSgenome.Dmelanogaster.UCSC.dm6"
## [26] "BSgenome.Drerio.UCSC.danRer5"
## [27] "BSgenome.Drerio.UCSC.danRer5.masked"
## [28] "BSgenome.Drerio.UCSC.danRer6"
## [29] "BSgenome.Drerio.UCSC.danRer6.masked"
## [30] "BSgenome.Drerio.UCSC.danRer7"
## [31] "BSgenome.Drerio.UCSC.danRer7.masked"
## [32] "BSgenome.Ecoli.NCBI.20080805"
## [33] "BSgenome.Gaculeatus.UCSC.gasAcu1"
## [34] "BSgenome.Gaculeatus.UCSC.gasAcu1.masked"
## [35] "BSgenome.Ggallus.UCSC.galGal3"
## [36] "BSgenome.Ggallus.UCSC.galGal3.masked"
## [37] "BSgenome.Ggallus.UCSC.galGal4"
## [38] "BSgenome.Ggallus.UCSC.galGal4.masked"
## [39] "BSgenome.Hsapiens.NCBI.GRCh38"
## [40] "BSgenome.Hsapiens.UCSC.hg17"
## [41] "BSgenome.Hsapiens.UCSC.hg17.masked"
## [42] "BSgenome.Hsapiens.UCSC.hg18"
## [43] "BSgenome.Hsapiens.UCSC.hg18.masked"
## [44] "BSgenome.Hsapiens.UCSC.hg19"
## [45] "BSgenome.Hsapiens.UCSC.hg19.masked"
## [46] "BSgenome.Hsapiens.UCSC.hg38"
```

```
## [47] "BSgenome.Hsapiens.UCSC.hg38.masked"
## [48] "BSgenome.Mfascicularis.NCBI.5.0"
## [49] "BSgenome.Mfuro.UCSC.musFur1"
## [50] "BSgenome.Mmulatta.UCSC.rheMac2"
## [51] "BSgenome.Mmulatta.UCSC.rheMac2.masked"
## [52] "BSgenome.Mmulatta.UCSC.rheMac3"
## [53] "BSgenome.Mmulatta.UCSC.rheMac3.masked"
## [54] "BSgenome.Mmusculus.UCSC.mm10"
## [55] "BSgenome.Mmusculus.UCSC.mm10.masked"
## [56] "BSgenome.Mmusculus.UCSC.mm8"
## [57] "BSgenome.Mmusculus.UCSC.mm8.masked"
## [58] "BSgenome.Mmusculus.UCSC.mm9"
## [59] "BSgenome.Mmusculus.UCSC.mm9.masked"
## [60] "BSgenome.Osativa.MSU.MSU7"
## [61] "BSgenome.Ptroglodytes.UCSC.panTro2"
## [62] "BSgenome.Ptroglodytes.UCSC.panTro2.masked"
## [63] "BSgenome.Ptroglodytes.UCSC.panTro3"
## [64] "BSgenome.Ptroglodytes.UCSC.panTro3.masked"
## [65] "BSgenome.Rnorvegicus.UCSC.rn4"
## [66] "BSgenome.Rnorvegicus.UCSC.rn4.masked"
## [67] "BSgenome.Rnorvegicus.UCSC.rn5"
## [68] "BSgenome.Rnorvegicus.UCSC.rn5.masked"
## [69] "BSgenome.Rnorvegicus.UCSC.rn6"
## [70] "BSgenome.Scerevisiae.UCSC.sacCer1"
## [71] "BSgenome.Scerevisiae.UCSC.sacCer2"
## [72] "BSgenome.Scerevisiae.UCSC.sacCer3"
## [73] "BSgenome.Sscrofa.UCSC.susScr3"
## [74] "BSgenome.Sscrofa.UCSC.susScr3.masked"
## [75] "BSgenome.Tgondii.ToxoDB.7.0"
## [76] "BSgenome.Tguttata.UCSC.taeGut1"
## [77] "BSgenome.Tguttata.UCSC.taeGut1.masked"
```

```
library("BSgenome.Scerevisiae.UCSC.sacCer2")
Scerevisiae
```

```
## Yeast genome:
## # organism: Saccharomyces cerevisiae (Yeast)
## # provider: UCSC
## # provider version: sacCer2
## # release date: June 2008
## # release name: SGD June 2008 sequence
## # 18 sequences:
## #   chrI    chrII   chrIII  chrIV   chrV    chrVI   chrVII  chrVIII c
## #   chrX    chrXI   chrXII  chrXIII chrXIV  chrXV   chrXVI  chrM    2
## # (use 'seqnames()' to see all the sequence names, use the '$' or '[[
## # operator to access a given sequence)
```

A *BSgenome* package contains a single object which is the second component of the name. At first, nothing is loaded into memory, which makes it very fast. You can get the length and names of the chromosomes without actually loading them.

```
seqlengths(Scerevisiae)
```

```
##    chrI   chrII  chrIII   chrIV    chrV   chrVI  chrVII chrVIII   chr
##  230208  813178  316617 1531919  576869  270148 1090947  562643  4398
##    chrX   chrXI  chrXII chrXIII  chrXIV   chrXV  chrXVI    chrM 2micr
##  745742  666454 1078175  924429  784333 1091289  948062   85779    63
```

```
seqnames(Scerevisiae)
```

```
##  [1] "chrI"    "chrII"   "chrIII"  "chrIV"   "chrV"    "chrVI"   "chr
##  [8] "chrVIII" "chrIX"   "chrX"    "chrXI"   "chrXII"  "chrXIII" "chr
## [15] "chrXV"   "chrXVI"  "chrM"    "2micron"
```

We load a chromosome by using the `[[` or `$` operators:

```
Scerevisiae$chrI
```

```
##   230208-letter "DNAString" instance
## seq: CCACACCACACCCACACACCCACACACCACACCA...GTGTGGGTGTGGTGTGGGTGTGGTGTG
```

We can now do things like compute the GC content of the first chromosome

```
letterFrequency(Scerevisiae$chrI, "CG", as.prob = TRUE)
```

```
##       C|G
## 0.3927361
```

To iterate over chromosomes seems straightforward with `lapply`. However, this function may end up using a lot of memory because the entire genome is loaded. Instead there is the `bsapply` function which handles loading and unloading of different chromosomes. The interface to `bsapply` is weird at first; you set up a `BSparams` object which contains which function you are using and which genome you are using it on (and a bit more information). This paradigm is being used in other packages these days, for example *BiocParallel*. An example will make this clear:

```
param <- new("BSParams", X = Scerevisiae, FUN = letterFrequency)
head(bsapply(param, letters = "GC"))
```

```
## $chrI
##   G|C
## 90411
##
## $chrII
##    G|C
## 311807
##
## $chrIII
##    G|C
## 121998
##
## $chrIV
##    G|C
## 580699
##
## $chrV
##    G|C
## 222141
##
## $chrVI
##    G|C
## 104636
```

note how the additional argument `letters` to the `letterFrequency` function is given as an argument to `bsapply`, not to the `BSParams` object. This gives us a list; you can simplify the output (like the difference between `lapply` and `sapply`) by

```
param <- new("BSParams", X = Scerevisiae, FUN = letterFrequency, simplif
bsapply(param, letters = "GC")
```

```
##    chrI.G|C   chrII.G|C  chrIII.G|C   chrIV.G|C    chrV.G|C   chrVI.G
##       90411      311807      121998      580699      222141      1046
##  chrVII.G|C chrVIII.G|C   chrIX.G|C    chrX.G|C   chrXI.G|C  chrXII.G
##      415227      216586      171122      286167      253728      4148
## chrXIII.G|C  chrXIV.G|C   chrXV.G|C  chrXVI.G|C    chrM.G|C 2micron.G
##      353167      303042      416443      360871       14676        24
```

Note how the mitochondria chromosome is very different. To conclude, the GC percentage of the genome is

```
sum(bsapply(param, letters = "GC")) / sum(seqlengths(Scerevisiae))
```

```
## [1] 0.3814872
```

SessionInfo

```
## R version 3.2.1 (2015-06-18)
## Platform: x86_64-apple-darwin13.4.0 (64-bit)
## Running under: OS X 10.10.5 (Yosemite)
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## attached base packages:
## [1] stats4    parallel  methods   stats     graphics  grDevices utils
## [8] datasets  base
##
## other attached packages:
##  [1] BSgenome.Scerevisiae.UCSC.sacCer2_1.4.0
##  [2] BSgenome_1.36.3
##  [3] rtracklayer_1.28.8
##  [4] Biostrings_2.36.3
##  [5] XVector_0.8.0
##  [6] GenomicRanges_1.20.5
##  [7] GenomeInfoDb_1.4.2
##  [8] IRanges_2.2.7
##  [9] S4Vectors_0.6.3
## [10] BiocGenerics_0.14.0
## [11] BiocStyle_1.6.0
## [12] rmarkdown_0.7
##
## loaded via a namespace (and not attached):
##  [1] knitr_1.11              magrittr_1.5
##  [3] zlibbioc_1.14.0         GenomicAlignments_1.4.1
##  [5] BiocParallel_1.2.20     stringr_1.0.0
##  [7] tools_3.2.1             lambda.r_1.1.7
##  [9] futile.logger_1.4.1     htmltools_0.2.6
## [11] yaml_2.1.13             digest_0.6.8
## [13] formatR_1.2             futile.options_1.0.0
## [15] bitops_1.0-6            RCurl_1.95-4.7
## [17] evaluate_0.7.2          stringi_0.5-5
## [19] BiocInstaller_1.18.4    Rsamtools_1.20.4
## [21] XML_3.98-1.3
```